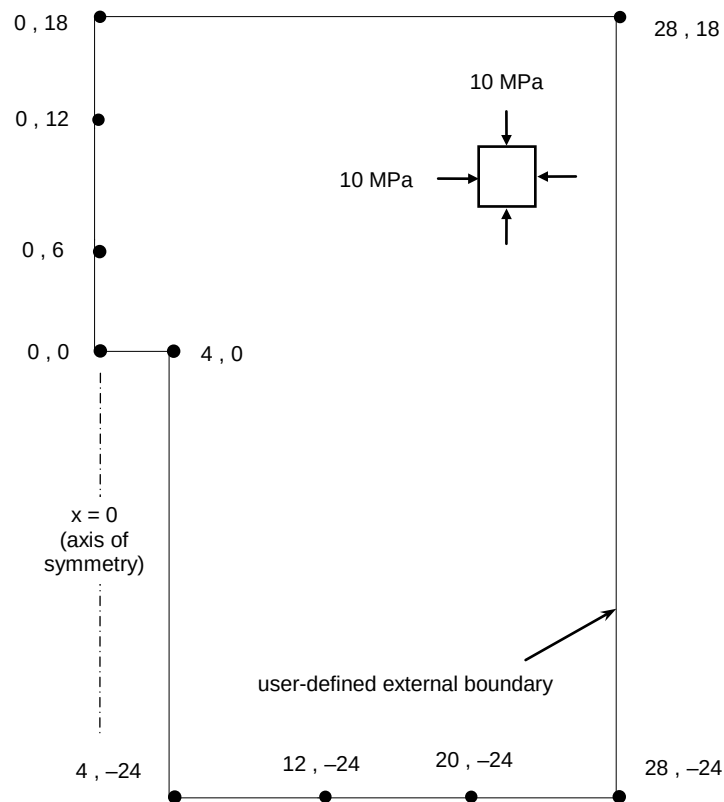


Axisymmetry Tutorial



This tutorial will illustrate the axisymmetric modeling option of *Phase2*. Axisymmetric modeling allows you to analyze a 3-D excavation which is rotationally symmetric about an axis. The input is 2-dimensional, but the analysis results apply to the 3-dimensional problem.

An Axisymmetric model in *Phase2* is typically used to analyze the end of a circular (or nearly circular) tunnel. The model we will be analyzing, shown above, represents the end of a cylindrical tunnel of 4 meter radius.

The finished product of this tutorial can be found in the **Tutorial 06 Axisymmetric.fez** file. All tutorial files installed with *Phase2* 8.0 can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

A few representations of simple axisymmetric models are shown below.

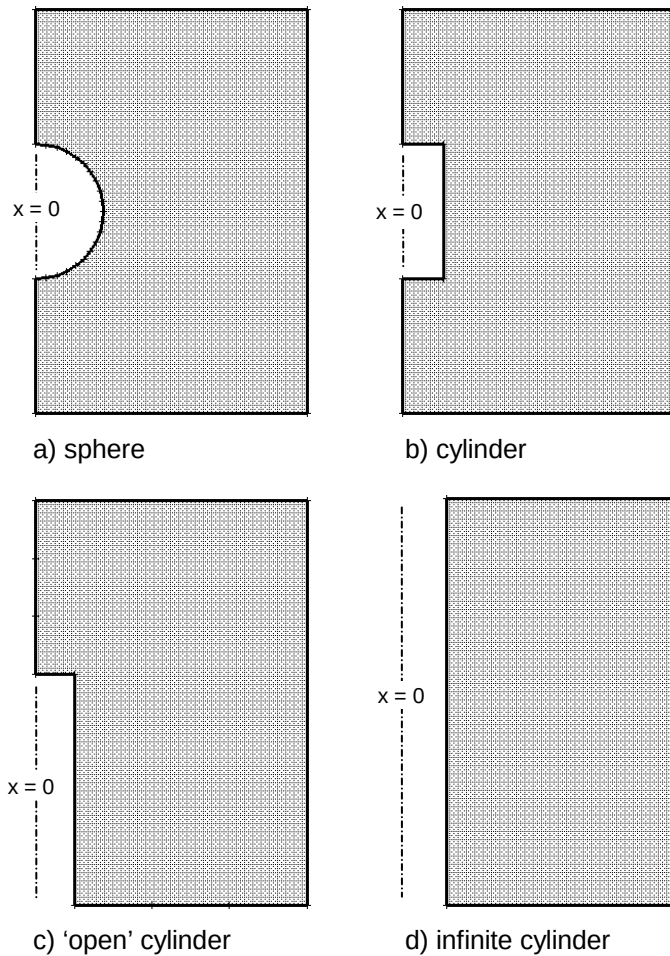


Figure 6-1: Simple axisymmetric models. For a), b) and c), the left edge of each boundary is coincident with the $X = 0$ (vertical) axis. For d), the boundary is displaced from the $X = 0$ axis, therefore modeling an infinite circular tunnel.

NOTE:

- Only an external boundary is necessary to define an axisymmetric model – the excavation is implicitly defined by the shape and location (relative to the $x=0$ axis) of the external boundary. Appropriate boundary conditions must also be applied to complete the modeling.
- The axis of rotation is always the $X = 0$ (vertical) axis. Your model must always be mapped to fit this convention, regardless of the actual orientation of the excavation. Because of the symmetry, only “half” of the problem needs to be defined.

So, to visualize an axisymmetric excavation, just imagine the shape formed by rotating the external boundary about the $x=0$ axis. Note that Figure 6-1 is for illustration, and that actually the boundaries should be extended relative to the excavations (in Figures 6-1a and 6-1b) to ensure that the fixed boundary conditions do not affect the results around the excavation. Figure 6-1d can actually be defined by a narrow horizontal strip, since the problem is effectively one-dimensional (i.e. results will only vary along a line perpendicular to the tunnel), and is in fact equivalent to a circular excavation in a plane strain analysis.

There are various restrictions on the use of axisymmetric modeling in *Phase2*, for example:

- the field stress must be axisymmetric i.e., aligned in the axial and radial directions.
- cannot be used with BOLTS (however LINERS are permitted)
- cannot be used with JOINTS
- all materials must have ISOTROPIC elastic properties

In this tutorial, we will look at results not only around the end of the tunnel, but also along its length, where the conditions are effectively plane strain. We will later verify these results by comparing with a plane strain analysis.

Model

Start the *Phase2* Model program.

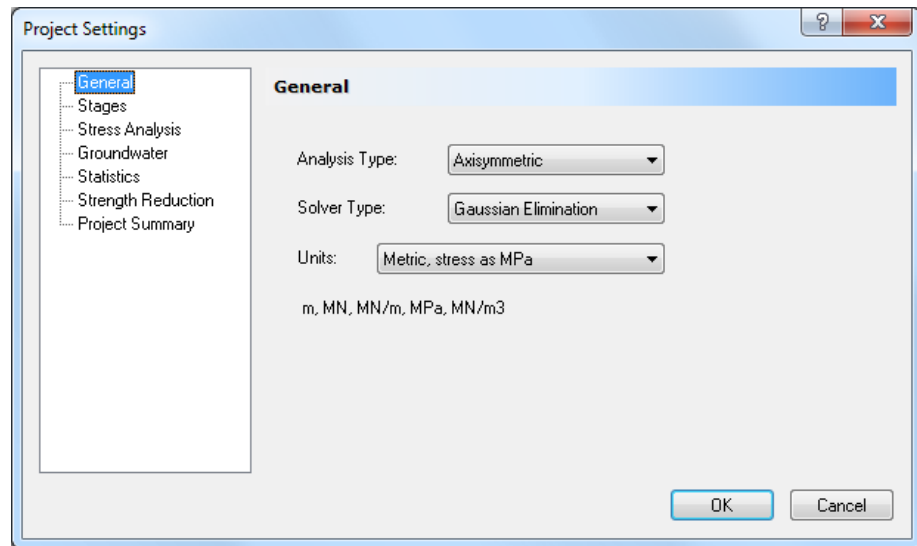
Project Settings

When you are creating an axisymmetric model, the first thing you should always do is set the Analysis Type to Axisymmetric in the Project Settings dialog.



Select: Analysis → Project Settings

- ✓ Enter:
 - Project Name = (optional)
 - Number of Stages = 1
- ✓ Analysis = Axisymmetric
 - Max. # of iterations = 500
 - Tolerance = 0.001
 - # Load Steps = Auto
 - Solver Type = Gauss. Elim.
- ✓ Units = Metric (MPa)



In the Project Settings dialog, toggle the Analysis Type to Axisymmetric, and select OK.

Entering Boundaries

Since only an external boundary is required to define an axisymmetric problem in *Phase2*, proceed directly to the Add External option (rather than the usual procedure of first adding excavations).



Select: Boundaries → Add External

Enter the following coordinates at the prompts:

```
Enter vertex [t=table,i=circle,esc=cancel]: 28 18
Enter vertex [...]: 0 18
Enter vertex [...]: 0 12
Enter vertex [...]: 0 6
Enter vertex [...]: 0 0
Enter vertex [...]: 4 0
Enter vertex [...]: 4 -24
Enter vertex [...]: 12 -24
```

```
Enter vertex [...]: 20 -24
Enter vertex [...]: 28 -24
Enter vertex [...,c=close,esc=cancel]: c
```

Select Zoom All (or press F2) to zoom the model to the center of the view.

This is the only boundary required for the problem, so we can proceed to the meshing.

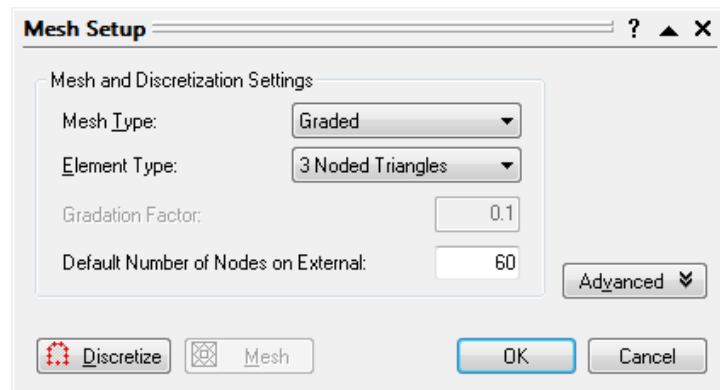
Meshing

As usual, we will discretize and mesh the model. However, let's first take a look at the Mesh Setup option.



Select: Mesh → Setup

✓ Enter:
Mesh Type = Graded
Elem. Type = 3 Noded Tri.
External Nodes = 60



Notice that the Mesh Setup dialog normally asks you for the # Excavation Nodes. However, for models which have no explicitly defined Excavation boundaries (such as this one), the # External Boundary Nodes is entered instead. Also, the Gradation Factor is not applicable when there are no Excavation boundaries defined. Select OK or Cancel, since we are using the default parameters.

Now let's discretize the external boundary.



Select: Mesh → Discretize

The status bar will indicate the actual number of discretizations created on the external boundary.

```
Discretizations: External=60
```

Note that this is a fairly coarse discretization. The boundary segments which are part of, or adjacent to, the excavation, will require a finer discretization. We can do this with the Custom Discretize option.

Custom Discretization



Select: Mesh → Custom Discretize

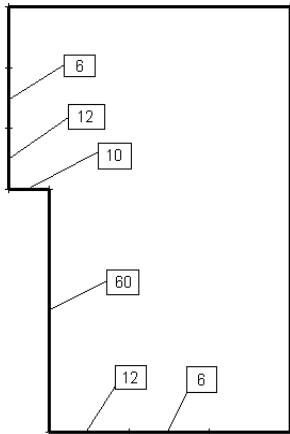
Select Segments to Discretize [enter=done,esc=cancel]: **use the mouse to select the long edge of the tunnel ie. the long vertical segment at the lower left of the external boundary. Right-click and select Done Selection, or just press Enter.**

In the Custom Discretize dialog, enter 60 as the number of discretizations, and select OK.

The length of the tunnel is now discretized into 60 elements. Now follow this same procedure to apply custom discretizations as indicated in the margin figure.

Note:

- you can select more than one line segment at a time, if they require the same number of discretizations (for example, the segments with 6 and 12 discretizations, in this case).
- the segments which are *not* marked in the margin figure, are to be left at their original discretizations.



CUSTOM DISCRETIZATION

Mesh

Now select the Mesh option from the toolbar or the Mesh menu, and the mesh will be generated, based on the discretization you just created.

Select: Mesh → Mesh

The status bar will indicate the total number of elements and nodes in the mesh.

NODES = 397 ELEMENTS = 648

At this point, we will make the following observation – you may have wondered, when we created the external boundary, why we added the extra vertices on the upper left vertical segment and lower right horizontal segment of the boundary, since these boundaries could have been defined by single segments. As you can now see, the extra segments allowed us to custom discretize the boundaries, in order to get a smooth transition between the fine mesh around the tunnel, and the coarser mesh of the rest of the boundary. (If we did not do this, a poor mesh would be generated where the fine to coarse transition is too abrupt.)



Boundary Conditions

In most of the tutorials so far, we have not been specifying any boundary conditions. We were using the default boundary condition, which is a fixed (zero displacement) external boundary.

For an axisymmetric model, the external boundary conditions are very important, and must be user specified. We cannot simply leave the boundary fixed, or else nothing would happen (i.e., no displacements could take place).

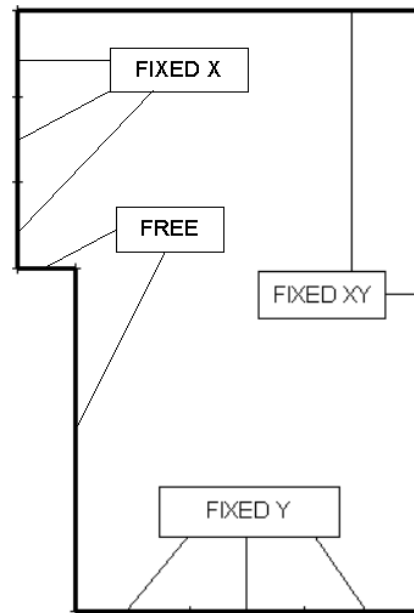


Figure 6-2: Displacement boundary conditions for axisymmetric model.

First, let's 'free' the tunnel boundaries.



Select: Displacements → Free

Select boundary segments to free [enter=done,esc=cancel]: **Use the mouse to select the 2 segments marked FREE in Figure 6-2. When finished, right-click and select Done Selection, or press Enter.**

The triangular pin symbols are now gone from the two boundary segments (representing the end of the tunnel and the length of the tunnel), indicating that they are free to move with no restriction in any direction.

Now let's specify the boundary segments at the upper left edge as restrained in the X direction, but free to move in the Y direction. (These segments are located on the axis of symmetry, and therefore must have zero X displacement).



Select: Displacements → Restrain X

Select boundary segments to restrain in X direction
[enter=done,esc=cancel]: **Use the mouse to select the three segments marked *FIXED X* in Figure 6-2. Right-click and select Done Selection, or press Enter.**

Observe that the triangular pins on these segments have been replaced by vertical rollers.

Now let's specify the boundary segments along the bottom as restrained in the Y direction, but free to move in the X direction.



Select: Displacements → Restrain Y

Select boundary segments to restrain in Y direction
[enter=done,esc=cancel]: **Use the mouse to select the three segments marked *FIXED Y* in Figure 6-2. Right-click and select Done Selection, or press Enter.**

Observe that the triangular pins on the bottom segments have been replaced by horizontal rollers.

Now we have a bit of tidying up to do.



Select: Displacements → Restrain X,Y

1. Right-click the mouse and select Pick by Boundary Nodes from the popup menu. This will change the mode of restraint application from boundary segments to boundary nodes. (The mode can also be changed in the Selection Mode sub-menu of the Displacements menu).
2. Select the upper left corner of the model, i.e. the vertex at (0 , 18).
3. Select the lower right corner of the model, i.e. the vertex at (28 , -24).
4. Right-click and select Done Selection.
5. A triangular pin symbol should now replace the roller symbol at these two vertices.

The above steps were necessary, since the upper left and lower right vertices required a Restrained XY condition. This leads us to an important point – after applying restraints to boundary segments, you should always check that nodes at the **ends** of segments have the correct conditions applied.

TIP: restraints can also be applied directly with a right-click shortcut, by right-clicking on segments or vertices and selecting a restraint option from the popup menu.

Field Stress

For this example, we will be using the default hydrostatic stress field of 10 MPa. However, let's look at the field stress option, to see how an axisymmetric field stress is specified.



Select: Loading → Field Stress

Only two independent principal stresses (Horizontal and Vertical) are specified for an axisymmetric problem, and no angle is allowed. Select OK or Cancel.

Note the following correspondences between Plane Strain and Axisymmetric field stress, as defined in *Phase2*:

PLANE STRAIN	AXISYMMETRIC
Sigma 1 (in-plane)	'Horizontal' stress
Sigma 3 (in-plane)	'Horizontal' stress
Sigma Z (out-of-plane)	'Vertical' stress
Angle	not applicable

Table 6-1: Equivalent plane strain and axisymmetric field stress components.

NOTE:

- The Horizontal (axisymmetric) field stress can also be thought of as a uniform radial stress around the excavation.
- An angle cannot be specified for the axisymmetric field stress, because this would require a true 3-dimensional analysis, which is beyond the scope of the *Phase2* axisymmetric analysis.

- It should be emphasized that the terms Horizontal and Vertical refer strictly to the setup of your model in *Phase2*, and not necessarily the true orientation of your excavation. The Vertical stress is the stress in the axial direction (i.e. the axis of rotational symmetry), and the Horizontal stress is the field stress perpendicular to this axis.

Properties

✓ Enter:

Name = Material 1
 Init.El.Ld.=Fld Stress Only
 Material Type = Isotropic
 Young's Modulus = 20000
 Poisson's Ratio = 0.3
 Failure Crit. = Mohr Coul.
 Material Type = Elastic
 Tens. Strength = 0
 Fric. Angle (peak) = 35
 Cohesion (peak) = 10.5

In this tutorial, we will not be defining or assigning any properties, therefore all default properties will be in effect. We have dealt with defining and assigning properties in previous tutorials.

For reference purposes, the default rock properties which will be in effect are shown in the margin. (If you want, you can verify this by selecting Properties→ Define Materials).

Note that our analysis will therefore be elastic. Also note the values of Young's Modulus and Poisson's ratio.

We have now completed the modeling. Your finished model should appear as shown below.

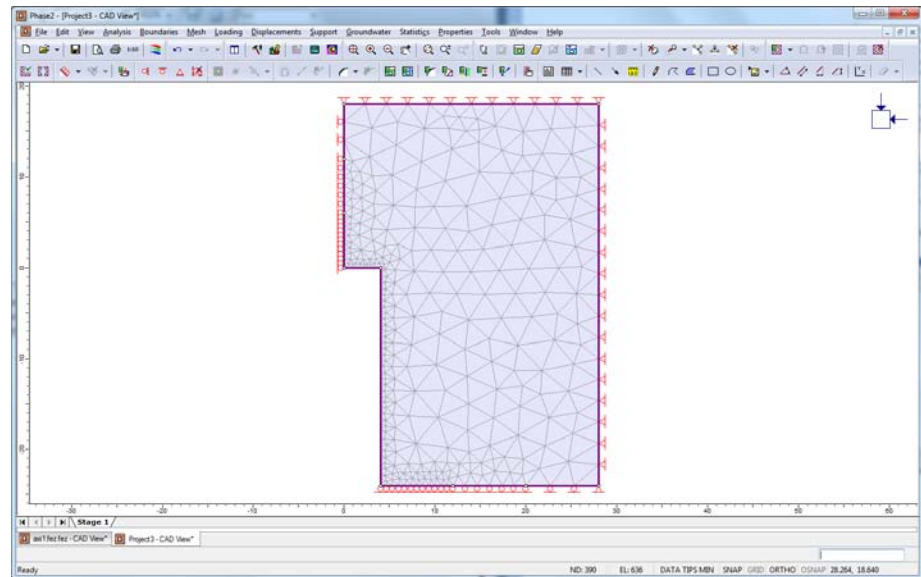


Figure 6-3: Finished model – *Phase2* Axisymmetric Tutorial

Compute

Before you analyze your model, save it as a file called **axi1.fez**.



Select: File → Save

Use the Save As dialog to save the file. You are now ready to run the analysis.



Select: Analysis → Compute

The *Phase2* Compute engine will proceed in running the analysis. When completed, you will be ready to view the results in Interpret.

Interpret

To view the results of the analysis:



Select: Analysis → Interpret

This will start the *Phase2* Interpret program.

Sigma 1

On the Sigma 1 contours, notice the stress concentration at the ‘corner’ of the tunnel (remember the tunnel is circular).



Toggle the stress trajectories on by selecting the Stress Trajectories button in the toolbar. The principal stress trajectories illustrate the “stress flow” around the end of the tunnel.

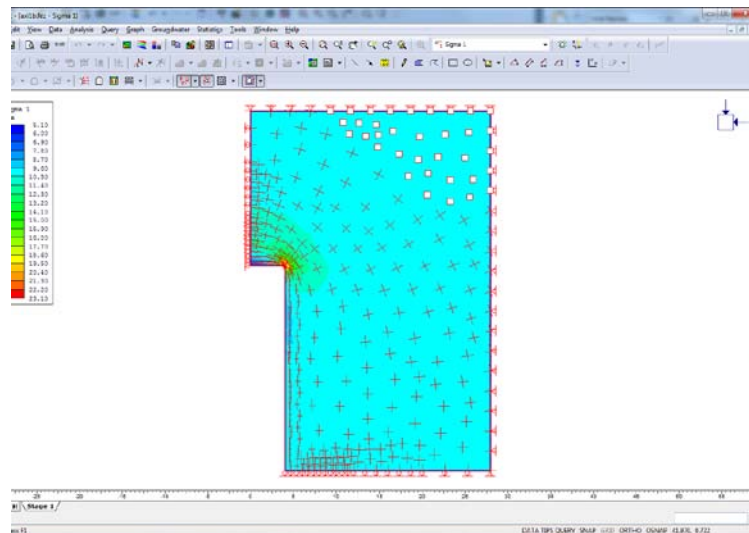


Figure 6-4: Principal stress trajectories around axisymmetric excavation.

The square dot markers in the upper right corner of the model indicate nodes where the difference between Sigma 1 and Sigma 3 is less than a certain tolerance, so that the conditions are effectively hydrostatic, and a distinction between ‘major’ and ‘minor’ principal stress is not warranted.

Toggle the stress trajectories off by re-selecting the Stress Trajectories toolbar button.

As an optional step, look at Sigma 3 and Sigma Z, and consider the significance of the principal stress results from an axisymmetric analysis. As with plane strain, Sigma 1 and Sigma 3 are the major and minor ‘in-plane’ principal stresses. Sigma Z is therefore the ‘out-of-plane’ stress, however, since the problem is axisymmetric, Sigma Z is really the induced circumferential or hoop stress around the excavation.

Displacement

Now let’s view the displacements.

Select:  **Total Displacement** ▼

Note the maximum total displacement displayed in the status bar is 0.00246 m, or just over 2 mm. Although this is quite small, remember that our analysis was elastic and we used a relatively high Young’s modulus.

Now let’s view the deformation vectors. Right-click the mouse and select Display Options.

In the Display Options dialog, toggle Deformation Vectors on, enter a Scale Factor of 600, and select Done.

The deformation vectors show the inward displacement along the length and face of the tunnel. Notice how the “corner” of the tunnel effectively restrains the displacements in both x and y directions.

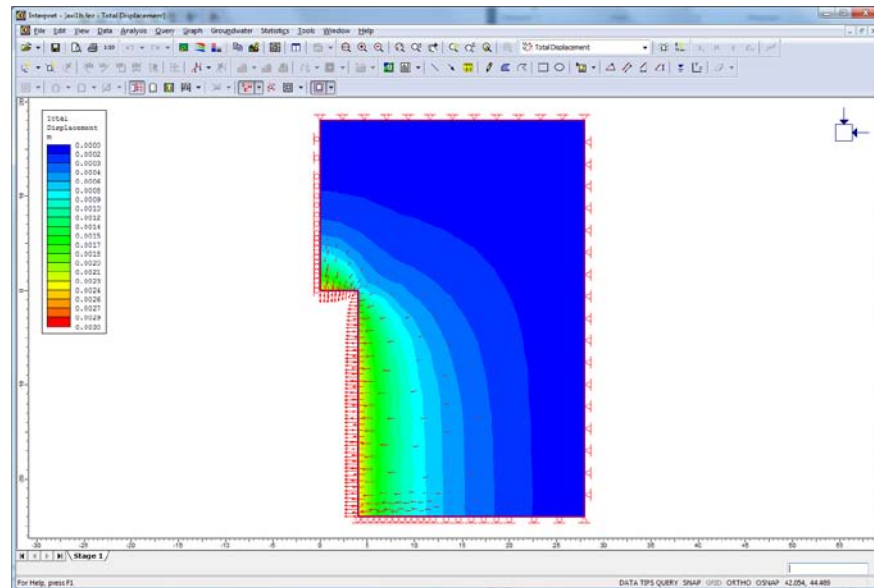


Figure 6-5: Displacement contours and vectors around axisymmetric excavation.

Toggle the deformation vectors off by selecting the Deformation Vectors button in the toolbar.

Query Data

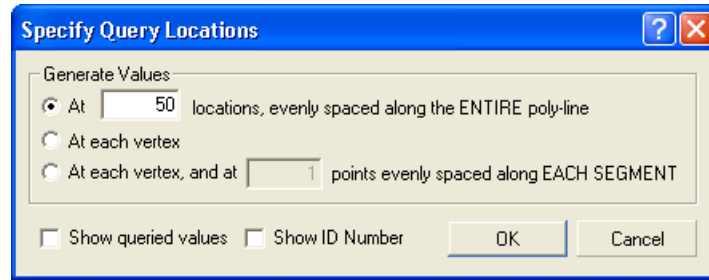
Phase2 allows the user to query data anywhere in the rock mass, to obtain values interpolated from the contour plots. A query can be a single point, a line segment, or an arbitrary polyline.

Let's first create a query along the length of the tunnel.



Select: Query → Add Material Query

1. It will be handy to use the Snap option, so right-click the mouse and make sure the Snap option is selected. While in Snap mode, if you place the cursor near a vertex, you can snap exactly to the location of a vertex.
2. Use the mouse to select the vertex at (4 , 0).
3. Use the mouse to select the vertex at (4 , -24).
4. Right-click the mouse and select Done. You will see the following dialog:



Toggle off Display Queried Values and Show ID Number and select OK.

A query has now been created along the length of the tunnel. The 50 locations at which data will be generated are indicated by black cross markers.

Graphing a Query



Graphs are created from queries with the Graph Material Queries option. However, a convenient shortcut to graph data for a single query, is to simply right-click on a query and select Graph Data.

1. Right-click on the query you just created (i.e. anywhere along the length of the tunnel), and select Graph Data from the popup menu.
2. You will see the Graph Query Data dialog. Select the Plot button in this dialog.
3. You should see the graph in Figure 6-6. Since we were viewing the Total Displacement contours, we obtained a graph of total displacement vs. distance along the query. The data graphed always corresponds to the contoured data being viewed.

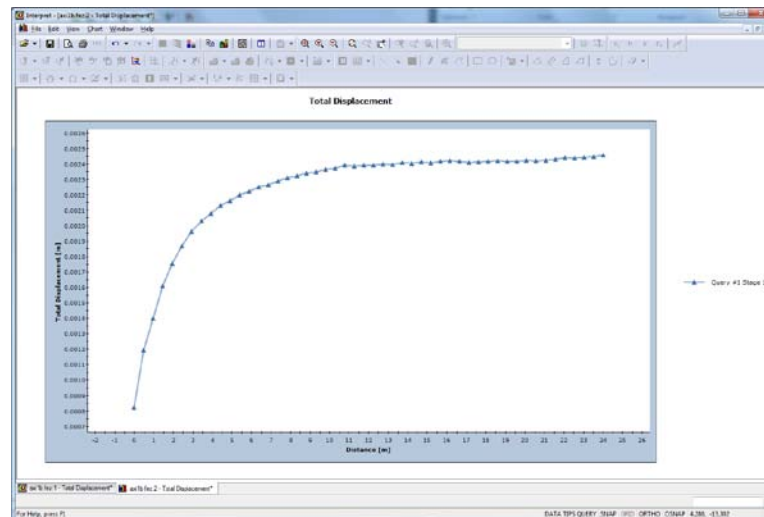


Figure 6-6: Total displacement along length of tunnel.

As you can see, the displacement levels off and becomes constant at a certain distance away from the tunnel face. This curve is useful in that it allows us to see the distance at which end effects can be ignored, and plane strain conditions can be assumed. Also, this curve can be used to estimate the “load split” factors, as described in the Support Tutorial, Step 2.

Deleting a Query



Queries are deleted with the Delete Material Query option. However, a convenient shortcut to delete a single query, is to simply right-click on the query and select Delete Query.

1. First close the graph if you are still viewing it.
2. Right-click on the query and select Delete Query, and the query will be removed from the model.

Graphing Multiple Queries

Now let's create two more queries, this time perpendicular to the tunnel, and plot them on the same graph.



Select: Query → Add Material Query

1. The Snap option should still be in effect, so use the mouse to select the vertex at (4 , -24), and then select the vertex at (28 , -24). Right-click the mouse and select Done.
2. In the Specify Query Locations dialog, enter 50 locations and select OK.

Add another query.



Select: Query → Add Material Query

1. Use the mouse to select the vertex at (4 , 0).
2. Now enter the coordinates (28 , 0) in the prompt line. Alternatively, you can right-click and select the Ortho snap option, which will allow you to snap the query exactly along a horizontal line, and also to the external boundary (at the point 28,0) because the Snap option is also enabled. Right-click the mouse and select Done.
3. In the Specify Query Locations dialog, enter 50 locations and select OK.

You have created two new queries, one along the lower edge of the model, and a parallel one at the face of the tunnel.

This time, to graph the queries, we will use the Graph Material Queries option, since we want both queries on the same graph. (The right-click shortcut can only be used to graph a single query).



Select: Graph → Graph Material Queries

1. Select the two queries by clicking on them with the left mouse button. (Alternatively, you could right-click the mouse and choose Select All from the popup menu.)
2. Right-click and select Graph Selected (or just press Enter), and you will see the Graph Query Data dialog.
3. Select the Plot button in the dialog, and you should see the following graph.

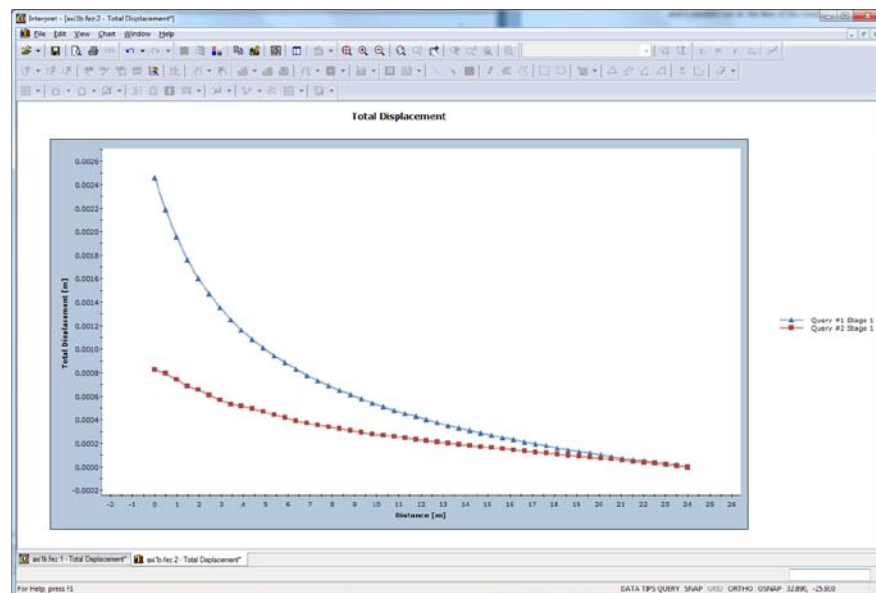


Figure 6-7: Total displacement perpendicular to tunnel, at face (lower curve), and at 24 meters from face (upper curve).

The total displacement decreases as we move away from the tunnel. Note:

- the Total Displacement along the lower boundary is exactly equivalent to the Horizontal (X) Displacement, since the Vertical (Y) Displacement along this boundary is zero. (If you graphed this query while viewing Horizontal Displacement instead of Total Displacement, you could verify this for yourself.)
- The Total Displacement curve at the face of the tunnel includes both X and Y displacement components.

Now close the graph view by selecting the X in the upper right corner of the view.

Plane Strain Comparison with Axisymmetric Results

To further illustrate the significance and meaning of an axisymmetric model, we will create a plane strain model which is equivalent in all respects to our axisymmetric model (except of course that the tunnel will now be infinite, with no 'end effects'), and compare the analysis results.

From Interpret, switch back to Model.



Select: Analysis → Model

(Or if you quit the program and are just re-starting the tutorial at this point, then start the *Phase2* Model program.)

Model

If you have been following this tutorial from the beginning, and the axisymmetric model is still open:



Select: File → New

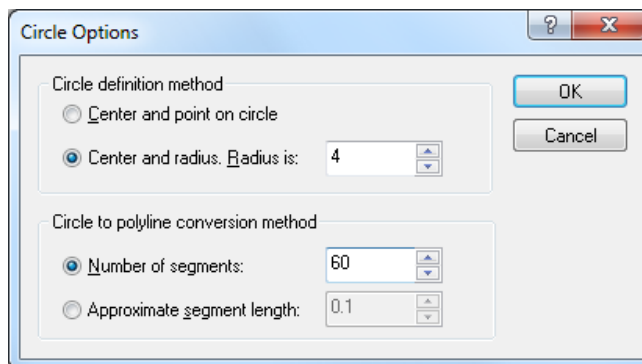
to open a new document window, so that you can begin creating a new model.

Plane strain is the default analysis type, so you do not have to set this in Project Settings, it will already be in effect. Let's first create a circular tunnel of radius 4 meters (i.e., the same radius as the axisymmetric tunnel).



Select: Boundaries → Add Excavation

1. Right-click and select the Circle option from the popup menu.
2. In the Circle Options dialog, select the Center and radius option, enter radius = 4, enter Number of segments = 60, and select OK.

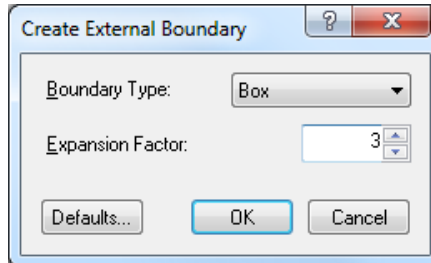


- At the prompt, enter 0,0 as the circle center, and the circular excavation will be created.

Now add the external boundary.



Select: Boundaries → Add External



We will use the default parameters, so just select OK to automatically create a BOX external boundary with an expansion factor of 3.

NOTE: this external boundary is the same distance away from the excavation, as the right edge of the external boundary for the axisymmetric problem (i.e. 28 meters from the center of the tunnel).

Mesh

Now discretize and mesh the model. First select Mesh Setup.



Select: Mesh → Setup

In the Mesh Setup dialog, set the Number of Excavation Nodes to 60. Select the Discretize button in the Mesh Setup dialog (this is equivalent to using the Discretize option in the Mesh menu). The status bar will indicate:

Discretizations: Excavation=60, External=68

Select the Mesh button in the Mesh Setup dialog (this is equivalent to using the Mesh option in the Mesh menu). The status bar will indicate:

NODES = 697 ELEMENTS = 1324

Select OK in the Mesh Setup dialog.

Boundary Conditions

We will use the default boundary condition, which is a fixed (i.e. zero displacement) condition on the external boundary. This corresponds to the Fixed XY condition of the *right edge* of the external boundary in the axisymmetric model.

Field Stress

We will use the default Field Stress (i.e. hydrostatic conditions $\sigma_1=\sigma_3=\sigma_z = 10\text{MPa}$, which is the same field stress we used for the axisymmetric problem), so you do not need to use the Field Stress option.

Properties

We will use the default rock material properties, so you do not have to enter or assign any properties. The default properties are Material Type = Elastic, Young's Modulus = 20,000 MPa, Poisson's ratio = 0.3.

However, we do have to excavate the tunnel. Let's do this with a right-click shortcut.

1. Right-click the mouse inside of the circular excavation.
2. In the popup menu, go to the Assign Material sub-menu, and select the Excavate option.
3. The circular tunnel is now excavated.

The model should appear as below.

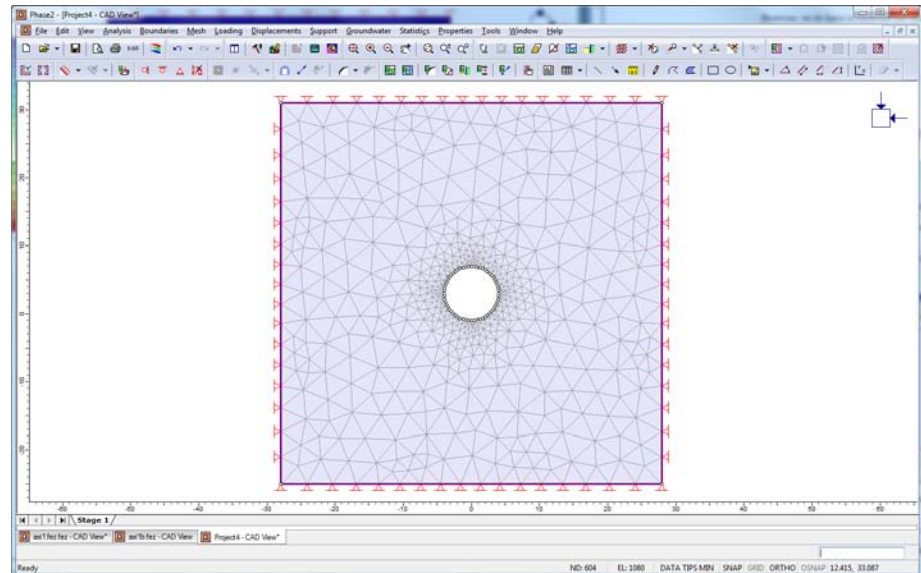


Figure 6-8: Infinite tunnel, 4 meter radius, plane strain problem.

Compute

Before you analyze your model, save it as a file called **axi2.fez**.



Select: File → Save

Use the Save As dialog to save the file. You are now ready to run the analysis. (Alternatively, if you select Compute before saving a new file, *Phase2* will recognize this, and display the Save As dialog).



Select: Analysis → Compute

The *Phase2* Compute engine will proceed in running the analysis. When completed, you will be ready to view the results in Interpret.

Interpret

We will now wrap up this tutorial with a few comparisons between the axisymmetric and plane strain models we have created. Switch back to Interpret.



Select: Analysis → Interpret

View the total displacement contours.

Select:  **Total Displacement**

Notice the maximum displacement displayed in the status bar.

Maximum Total Displacement = 0.002416 m

This is almost identical to the maximum displacement from the axisymmetric problem (0.00246).

Now let's use the Query and Graph options again to plot the displacement vs. distance from the tunnel boundary.



Select: Query → Add Material Query

1. Enter the point (4 , 0) at the first prompt and (28 , 0) at the second prompt. Right-click and select Done, or just press Enter.
2. In the Specify Query Locations dialog, enter 50 locations, and select OK.
3. Notice the query created from the right edge of the tunnel to the right edge of the external boundary. Right-click on the query and select Graph Data.
4. You will see the Graph Query Data dialog. Select the Plot button and a graph of total displacement will immediately be generated.

You can see the displacement decreasing from the maximum at the tunnel boundary, to zero at the fixed, external boundary.

Finally, let's compare this curve with the equivalent query from the axisymmetric analysis.

1. Open the axisymmetric model (**axi1.fez**) in the Interpret program.
2. Right-click on the lower query (at the bottom edge of the model) and select Graph Data. In the Graph Query Data dialog select the Plot button to generate the graph.
3. Now tile the views by selecting Tile Windows from the toolbar. Your screen should look similar to the following.

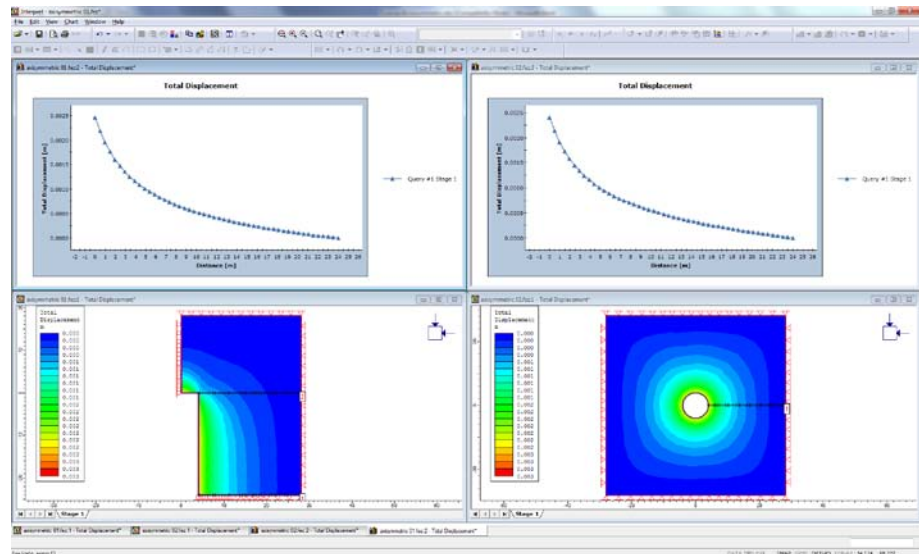


Figure 6-9: Comparison of displacements for axisymmetric and plane strain model.

The Total Displacement graphs from the two models are nearly identical.

- One graph represents the query along the lower edge of the axisymmetric model.
- The other is the query added on the plane strain model.

This illustrates the relationship between the axisymmetric and plane strain models – although the two models look very different, we can extract the same results from either one.

Additional Exercises

When you plotted the total displacement contours for the plane strain tunnel model, you may not have noticed, but the contours begin to get “square” as you get further from the tunnel (immediately around the tunnel they are circular). The displacements are conforming to the shape of the external boundary and the fixed boundary condition we imposed on it.

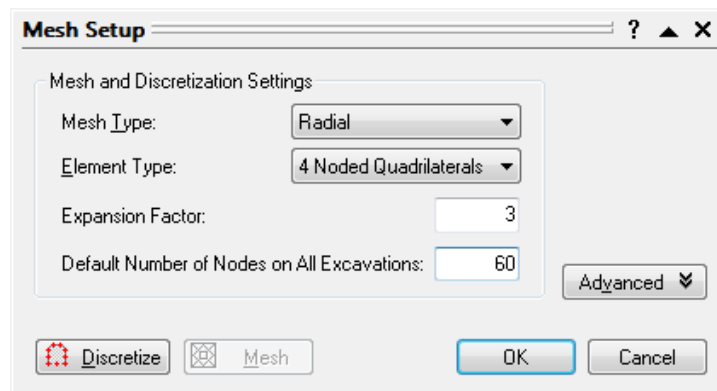
Radial Mesh

For circular problems such as this one, there is a more appropriate meshing option we could have used, called Radial meshing. Radial meshing produces a symmetric, reproducible radial mesh for symmetric problems such as this.

As an additional exercise, re-do the plane strain circular tunnel analysis, with the following changes:

- After you add the excavation, DO NOT add the external boundary, but select Mesh Setup instead.
- In the Mesh Setup dialog, toggle the Mesh Type to Radial, the Element Type to 4 Noded Quadrilaterals, and enter the #Excavation Nodes = 60. Note that for a Radial mesh, an Expansion Factor for the external boundary is entered, rather than a Gradation Factor.

- ✓ Enter:
 - ✓ Mesh Type = Radial
 - ✓ Elem. Type = 4 Node Quad
Expansion Factor = 3
 - ✓ # Excavation Nodes = 60



- Discretize and Mesh. The external boundary will appear when the radial mesh is generated.
- Carry out the analysis and data interpretation as before.

When you plot the displacement versus distance from the tunnel, you should get nearly identical results as when you used the BOX external boundary. However, the displacement contours are no longer distorted, and are circular at any distance from the excavation.

Also note that quadrilateral elements in conjunction with Radial meshing, are very efficient and give very good results.

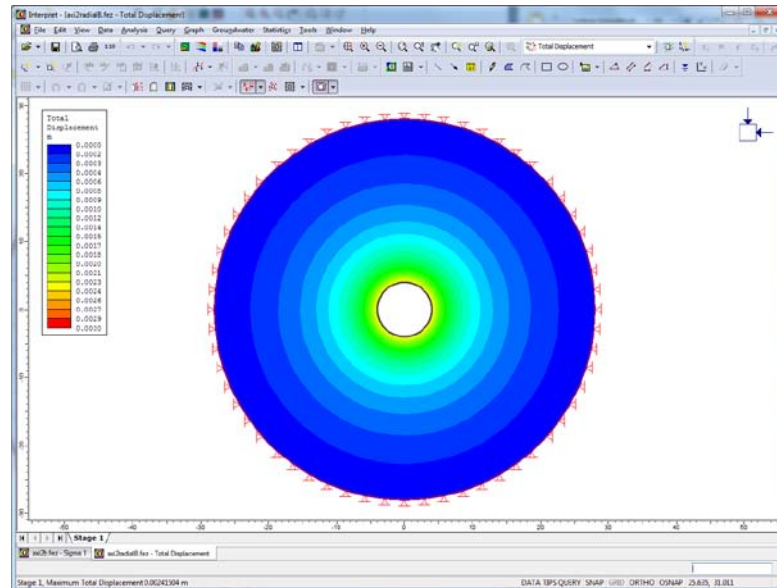


Figure 6-10: Total displacement contours with radial mesh.

Distance of External Boundary from Excavation

As pointed out in other tutorials, the distance of the external boundary from the excavation(s), and the boundary conditions we impose on it, are very important.

1. Create another Radial mesh, as described above, except this time use an Expansion Factor of 5 (in the Mesh Setup dialog).
2. Re-run the analysis and plot the displacement versus distance from the tunnel (add a query from 4,0 to 44,0).
3. Export the query data to Excel for both radial mesh models (expansion factor = 3 and expansion factor = 5). Note: you can export query data by right-clicking on a query and selecting Copy Data or right-click on a graph and select Plot in Excel.
4. In Excel plot the query data to obtain Figure 6-11.

You will see that our previous Expansion Factor of 3 was too low, because displacements increase significantly when we move the fixed external boundary farther from the tunnel.

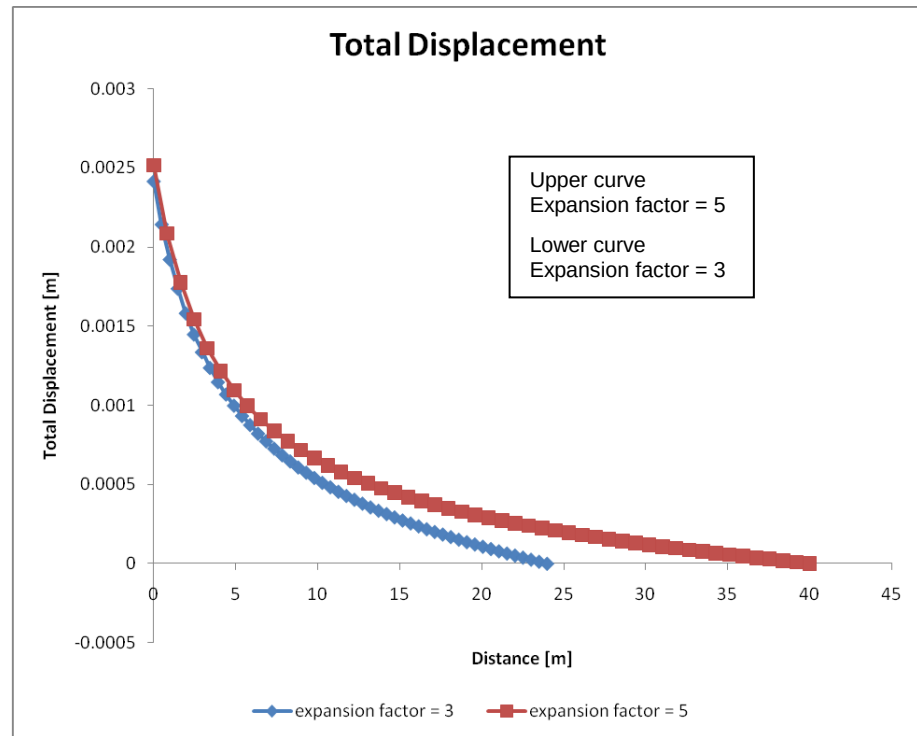


Figure 6-11: Moving the fixed external boundary farther from the excavation results in increased displacements.

The displacements near the excavation are comparable, but diverge towards the external boundary. For example, at about 18 meters from the excavation, the displacement for the Expansion Factor = 5 curve is about double the Expansion Factor = 3 curve (see Figure 6-11).

The restraining effect of a fixed external boundary should therefore always be considered. When comparing with analytical solutions, as in the *Phase2* verification manual, it is very important to take this into account.

One final suggested exercise:

Re-do the axisymmetric problem and move the right edge of the external boundary over to 44 meters. This gives an equivalent distance from the excavation as the plain strain model with an expansion factor of 5. Compare results with the equivalent plain strain model.